

SUPREME CAST

Leading Manufacturer of C.I and S.G.I Castings

**SUPREME
CAST**

“Crafting Excellence in Every Mould”

Our Vision

"Our vision is to excel as a premier company, specializing in the production of high-quality C.I and S.G iron machined components. We aim to be recognized globally for our innovation, precision, and commitment to meeting the diverse requirements of leading nations, thereby contributing significantly to industrial advancements and technological progress."

SUPREME
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Quality Policy

"At Supreme Cast, we are deeply committed to delivering products of the highest quality, knowing that they reflect the integrity of our processes. We adhere to industry best practices throughout our manufacturing operations and offer flexibility for our customers to tailor processes to their unique needs. Our primary objective is to provide superior quality parts at competitive prices, driving our strategy to prioritize excellence in both cost-effectiveness and product quality. Our dedicated team of engineers continuously refines our manufacturing processes to ensure we remain at the forefront of the industry in terms of both cost efficiency and product excellence."

CAST

History & Future

Inception (2018): Our journey began in 2018 with a commitment to excellence. The implementation of the GREEN Sand Process marked our inception, providing quality casting solutions with a manufacturing capacity of 150MT. This laid the foundation for our dedication to meeting the evolving requirements of our customers.

Expansion with Machine Shop Facility: Responding to the dynamic needs of the industry, we took a significant step forward by adding a dedicated Machine Shop Facility as a separate unit. This expansion emphasized our commitment to precision and comprehensive solutions, ensuring we meet and exceed customer expectations.

Foundry Unit-II (2019): In 2019, we further strengthened our capabilities by inaugurating Foundry Unit-II. Operating on the GREEN Sand Process, this unit elevated our manufacturing capacity to 200MT, demonstrating our continuous growth and adaptability to market demands.

History & Future

Current Milestone (June 2024): Looking ahead, we are on the brink of a transformative leap. By June 2024, we are set to introduce a large-scale manufacturing facility with all the required in-house capabilities. This strategic expansion is a testament to our commitment to delivering enhanced value and ensuring the fulfillment of our customers' diverse needs.

Future Vision (End of 2026): As part of our forward-thinking vision, we have a definitive plan to establish a cutting-edge No-Bake Molding Process by the end of 2026. This initiative is designed to produce substantial 200kg weight parts with an impressive manufacturing capacity of 2000MT. Our commitment to innovation and growth is unwavering, positioning us as industry leaders dedicated to serving our customers with excellence.

At **Supreme Cast**, we are not just in the business of manufacturing; we are committed to a journey of continuous growth, meeting customer requirements, and setting new standards in the industry. The future holds exciting possibilities as we evolve and adapt to the ever-changing landscape.

Infrastructure

We take pride in our robust infrastructure designed to meet the demands of precision casting and machining. Here's an overview of our key facilities:

Land Area: We operate on a sprawling land area of 1,25,000 sq.ft, providing ample space for various production and operational needs. This expansive landscape forms the foundation for our cutting-edge manufacturing processes.

Foundry Work Area: Within our premises, we have allocated 60,000 sq.ft specifically for our foundry operations. This dedicated space is optimized for the GREEN Sand Process, ensuring efficient and high-quality casting production.

Machine Shop Work Area: Complementing our foundry facilities, our Machine Shop Work Area spans 11,000 sq.ft. Here, advanced machining processes and precision engineering take place, guaranteeing the highest standards in the final product.

Our strategically planned infrastructure allows us to seamlessly integrate casting and machining operations, providing end-to-end solutions for our valued customers. The spacious work areas ensure a smooth workflow, facilitating precision and efficiency in every stage of our manufacturing process. As we look towards the future, we remain committed to enhancing and expanding our infrastructure to meet the evolving needs of our industry.

Our Molding Capacity & Benefits

1.High Production Capacity:

The use of six set jolter machines, each handling a different pattern simultaneously, contributes to a robust production capacity. Producing a minimum of 3000 molds in a single day showcases the efficiency and scalability of the operation.

2.Versatile Production with Multiple Patterns:

1. The ability to use six different patterns concurrently allows for versatile and diverse production. This is particularly beneficial for foundries dealing with a range of casting requirements, providing flexibility in meeting customer demands.

3.Large Mold Size Capability:

1. The jolter molding machines can accommodate a maximum box size of 560 mm X 560 mm X 75 to 350 mm. This capability enables the production of larger molds, expanding the scope of applications and catering to a variety of casting sizes.

4.Optimized Air Pressure for Mold Hardness:

1. Utilizing 0.5 Kg. air pressure ensures the proper hardness of the mold. This optimization is crucial in preventing common issues such as swelling, sand wash, and weight variation, resulting in high-quality and consistent castings.

5.Precise Mismatch Control:

1. With a maximum mismatch of 0.750 mm, the jolter molding machines deliver precise and accurate molds. This level of control minimizes the need for rework and adjustments, contributing to the overall quality of the castings.

6.Controlled Draft Angles: Maintaining a draft angle within 1.5 degrees facilitates smooth mold release. Controlled draft angles are essential for reducing the likelihood of defects, ensuring proper pattern withdrawal, and enhancing overall mold integrity.

7.High Molding Rates: Achieving a molding rate of 150 molds per hour without core casting and 120 molds per hour with core casting highlights the efficiency and speed of the jolter molding machines. This rapid production rate is crucial for meeting tight deadlines and large-scale production requirements.

Our Melting Capacity & Benefits:

- 1.Versatility in Production:** The availability of three induction furnaces with capacities of 500kg, 300kg, and 150kg allows for versatility in production. This enables the foundry to handle various casting sizes and adapt to diverse customer requirements.
- 2.Optimized Metal Melting:** The combination of different furnace capacities facilitates efficient and customized metal melting based on specific production needs. This flexibility minimizes energy consumption, ensuring that the appropriate furnace is used for the required metal volume.
- 3.High Melting Capacity:** The collective capacity of the three furnaces allows for a substantial melting capacity, with a minimum of 2500kgs of liquid metal per hour when all furnaces are operational. This high melting capacity contributes to increased production rates and throughput.
- 4.Energy-Efficient Power Supply:** The 800 KVA power supply ensures a stable and reliable source of energy to operate the induction furnaces. This level of electrical capacity supports energy-efficient melting processes, contributing to cost-effective and sustainable manufacturing.
- 5.Cost-Effective Production:** The ability to choose the appropriate furnace size for different metal quantities enhances cost-effective production. This results in minimized energy wastage and optimized operational efficiency, ultimately reducing production costs.
- 6.Adaptability to Various Metal Alloys:** Induction furnaces are well-suited for melting a variety of metal alloys. This adaptability is crucial for foundries dealing with diverse casting materials, allowing for flexibility in catering to different industry requirements.
- 7.Improved Safety and Ergonomics:** Induction furnaces offer improved safety features, including reduced exposure to harmful emissions and minimized physical strain on operators. This contributes to a safer and more comfortable working environment for foundry personnel.

Core Making And Capacity

1.PLC Controlled Cold Box Core Shooters:

1. These core shooters are equipped with programmable logic controllers (PLCs) for precise control over the cold box core shooting process. PLC technology enhances automation and accuracy in core production.

2.Pneumatic Sand Transporter and PLC Controlled Binder Dosing Units:

1. The inclusion of a pneumatic sand transporter and PLC-controlled binder dosing units ensures efficient and controlled delivery of sand and binders during the core-making process. This contributes to consistency and quality in core production.

•Cold Core Shooters: 5 Kg Cold Core Shooter (Quantity: 2)

- 10 Kg Cold Core Shooter (Quantity: 2)
- 20 Kg Cold Core Shooter (Quantity: 1)

These cold core shooters, ranging from 5 kg to 20 kg capacity, offer flexibility in producing cores of varying sizes using the cold box process.

1.Hot Core Shooter (Shell Core): 12 Kg Hot Core Shooter (Quantity: 2)

1. The 12 kg hot core shooter is specifically designed for creating shell cores using a heated process. This method is often employed for applications requiring higher core strengths and precise detailing.

Lab Testing Facility

- 1.Spectro Meter (Make: Metal Power):** Utilized for analyzing the chemical composition of metals. It provides accurate information about the elemental composition of materials.
- 2.CE-Meter (Make: Suyesh Solution P. Ltd.):** CE-Meter, or conductivity-electivity meter, is crucial for measuring the electrical conductivity and electivity of materials. This is particularly important for evaluating the purity and quality of substances.
- 3.Hardness Testing Machine (Make: Royal Enterprise):** Used to determine the hardness of materials. Hardness testing is vital for understanding the mechanical properties and suitability of materials for specific applications.
- 4.Sand Rammer (Make: Royal Enterprise):** Essential for compacting foundry sand in molds or cores. Sand ramming ensures proper molding and casting conditions.
- 5.Strength Machine (Make: Royal Enterprise):** Measures the compressive strength of materials. This information is crucial in assessing the load-bearing capacity of components.
- 6.Rapid Moisture Tester (Make: Royal Enterprise):** Determines the moisture content in materials quickly. This is vital for quality control in various manufacturing processes.
- 7.Permeability Tester (Make: Royal Enterprise):** Evaluates the permeability of molding sand. Permeability is a key factor in the efficiency of the molding process.
- 8.Clay Washer (Make: Royal Enterprise):** Used for washing clay from sand samples. This is important in preparing clean sand for molding purposes.
- 9.Mould Hardness Tester (Make: Royal Enterprise):** Measures the hardness of molds used in the foundry process. This helps ensure the durability and quality of molds.
- 10.Core Hardness Tester (Make: Royal Enterprise):** Evaluates the hardness of cores used in casting. Core hardness is critical for maintaining the integrity of castings

CNC Machine Facility:

- **Number of CNC Machines:** 4
- **Type of CNC Machines:** Turning
- **Turning Diameter:** 350mm
- **Swing Diameter:** 600mm
- **Make:** Macpower CNC Machines Limited, India

Key Points:

- 1.Machine Type:** The CNC machines in your facility are designed for turning operations, indicating precision in creating cylindrical components.
- 2.Quantity:** With four CNC machines, your facility has the capacity for simultaneous or sequential turning operations, allowing for efficient production.
- 3.Turning Diameter:** The specified turning diameter of 300mm indicates the maximum size of the components that can be machined on these CNC turning centers.
- 4.Swing Diameter:** The swing diameter of 500mm suggests the maximum diameter of the workpiece that can be accommodated while it rotates on the machine.
- 5.Make:** The machines are manufactured by Macpower CNC Machines Ltd., a reputable company based in Bangalore, India. Macpower CNC Machines is known for producing high-quality CNC machines.

VMC Machine Facility:

1. Machine 1:

1. **Type:** VMC Machine with Double Pallet Changer
2. **Work Envelope (Table Size):** 780mm X 450mm
3. **Additional Features:** Double Pallet Changer
4. **Make:** MACPOWER CNC MACHINES

Key Points:

1. Double Pallet Changer: The first VMC machine is equipped with a double pallet changer, allowing for continuous machining with reduced setup time. This feature enhances productivity and efficiency.

2. Work Envelope: The work envelope or table size of each machine (500mm X 450mm for the first machine and 800mm X 450mm for the second machine) specifies the maximum dimensions of the workpieces that can be accommodated.

Our Customers

- 1.Kirloskar Brothers Ltd. (Sanand)
- 2.Waterman Industries Pvt Ltd (Ahmedabad)
- 3.KSB Ltd. (Nasik)
- 4.Angel Pump Pvt. Ltd. (Rajkot)
- 5.Hayat Pumps Pvt Ltd (Rajkot)
- 6.Silver Pump Pvt. Ltd. (Rajkot)
- 7.Sabar Pumps Pvt. Ltd. (Ahmedabad)
- 8.Captain Tractors Pvt. Ltd. (Rajkot)
- 9.Pluga Pumps and Motors Pvt. Ltd. (Vadodara)
- 10.Shakti Pumps India Ltd. (Pithampur)
- 11.Tirth Agro Technology Private Limited. (Rajkot)
- 12.Mangla Engineering Limited (Dewas)
- 13.Ansons Electro Mechanical Works (Mumbai)
- 14.Shri Alka Industries (Vijapur)
- 15.Power Drive Bearing Pvt. Ltd. (Sanand)
- 16.Graphicarb Products (Ahmedabad)
- 17.Vikas Agro Pvt. Ltd. (Indore)
- 18.Puzzolana Machinery LLP (Hyderabad)
- 19.Unnati Pumps Pvt. Ltd. (Ahmedabad)
- 20.Maxtron Elemech (Ahmedabad)

Special Competitive Edge

At **SUPREME CAST**, we pride ourselves on delivering a unique competitive edge that goes beyond conventional offerings. We commit to providing our customers with a minimum annual savings of 3%, achieved through:

- 1.Minimum Cost of Development:** Our streamlined processes and cutting-edge technologies ensure that the development costs are kept to a minimum, allowing you to enjoy cost-effective solutions without compromising on quality.
- 2.Weight Reduction:** Through innovative design and material optimization, we focus on achieving weight reduction in our products. This not only enhances efficiency but also contributes to cost savings, particularly in shipping and operational aspects.
- 3.Operational Cost Reduction:** We are dedicated to identifying and implementing solutions that result in a reduction of operational costs for our customers. This encompasses a holistic approach, addressing various aspects of production and usage.
- 4.Key Manufacturing: Capabilities All Grey Grades Under One Roof** With a substantial monthly capacity of 800MT, we offer a comprehensive range of grey grades, providing you with a single-source solution for all your casting needs.
- 5.Versatile Weight Range:** Our manufacturing capabilities cover a weight spectrum from 1kg to 49kg for single-piece castings. This versatility allows us to cater to diverse industry requirements.
- 6.Problem-Solving Expertise:** Leveraging our expertise, we specialize in providing customized solutions to address your unique challenges. Consider us not just as a supplier but as a partner in overcoming obstacles and achieving optimal results.



THANK YOU
FOR
VISIT OUR PROFILE